

SEACROSS PEMETREXED POWDER FOR CONCENTRATE FOR SOLUTION FOR INFUSION

1. NAME OF THE MEDICINAL PRODUCT

Pemetrexed 100 mg powder for concentrate for solution for infusion

Pemetrexed 500 mg powder for concentrate for solution for infusion

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each vial contains 100 mg of pemetrexed (as pemetrexed disodium).

Each vial contains 500 mg of pemetrexed (as pemetrexed disodium).

After reconstitution, each vial contains 25 mg/ml of pemetrexed.

Excipients with known effect:

Each 100 mg vial contains less than 1 mmol (11 mg) of sodium.

Each 500 mg vial contains approximately 2.34 mmol (54 mg) of sodium.

3. PHARMACEUTICAL FORM

Powder for concentrate for solution for infusion.

White to either light yellow or green-yellow lyophilised powder.

After reconstitute: Clear, colorless to yellow(Y3) or yellow-green(GY3) colored solution.

Compatible Diluent:

sodium chloride 9mg/ml (0.9%) solution for injection

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Malignant pleural mesothelioma

Pemetrexed in combination with cisplatin is indicated for the treatment of chemotherapy naïve patients with unresectable malignant pleural mesothelioma.

Non-small cell lung cancer

Pemetrexed in combination with cisplatin is indicated for the first line treatment of patients with locally advanced or metastatic non-small cell lung cancer other than predominantly squamous cell histology.

Pemetrexed is indicated as monotherapy for the maintenance treatment of locally advanced or metastatic non-small cell lung cancer other than predominantly squamous cell histology in patients whose disease has not progressed immediately following platinum-based chemotherapy.

Pemetrexed is indicated as monotherapy for the second line treatment of patients with locally advanced or metastatic non-small cell lung cancer other than predominantly squamous cell histology.

Pemetrexed in combination with pembrolizumab and platinum chemotherapy, is indicated for the first line treatment of patients with metastatic non-squamous NSCLC, with no EGFR or

ALK genomic tumor aberrations

4.2 Posology and method of administration

Posology:

Pemetrexed must only be administered under the supervision of a physician qualified in the use of anti-cancer chemotherapy.

Pemetrexed in combination with cisplatin

The recommended dose of Pemetrexed is 500 mg/m^2 of body surface area (BSA) administered as an intravenous infusion over 10 minutes on the first day of each 21-day cycle. The recommended dose of cisplatin is 75 mg/m^2 BSA infused over two hours approximately 30 minutes after completion of the pemetrexed infusion on the first day of each 21-day cycle. Patients must receive adequate anti-emetic treatment and appropriate hydration prior to and/or after receiving cisplatin.

Pemetrexed as single agent

In patients treated for non-small cell lung cancer after prior chemotherapy, the recommended dose of pemetrexed is 500 mg/m^2 BSA administered as an intravenous infusion over 10 minutes on the first day of each 21-day cycle.

Premedication regimen

To reduce the incidence and severity of skin reactions, a corticosteroid should be given the day prior to, on the day of, and the day after pemetrexed administration. The corticosteroid should be equivalent to 4 mg of dexamethasone administered orally twice a day.

To reduce toxicity, patients treated with pemetrexed must also receive vitamin supplementation (see section 4.4). Patients must take oral folic acid or a multivitamin containing folic acid (350 to 1000 micrograms) on a daily basis. At least five doses of folic acid must be taken during the seven days preceding the first dose of pemetrexed, and dosing must continue during the full course of therapy and for 21 days after the last dose of pemetrexed. Patients must also receive an intramuscular injection of Vitamin B₁₂ (1000 micrograms) in the week preceding the first dose of pemetrexed and once every three cycles thereafter. Subsequent Vitamin B₁₂ injections may be given on the same day as pemetrexed.

Monitoring

Patients receiving pemetrexed should be monitored before each dose with a complete blood count, including a differential white cell count (WCC) and platelet count. Prior to each chemotherapy administration blood chemistry tests should be collected to evaluate renal and hepatic function. Before the start of any cycle of chemotherapy, patients are required to have the following: absolute neutrophil count (ANC) should be $\geq 1500 \text{ cells/mm}^3$ and platelets should be $\geq 100,000 \text{ cells/mm}^3$.

Creatinine clearance should be $\geq 45 \text{ ml/min}$.

The total bilirubin should be ≤ 1.5 times upper limit of normal. Alkaline phosphatase (AP), aspartate aminotransferase (AST or SGOT) and alanine aminotransferase (ALT or SGPT) should be ≤ 3 times upper limit of normal. Alkaline phosphatase, AST and ALT ≤ 5 times upper limit of normal is acceptable if liver has tumour involvement.

Dose adjustments

Dose adjustments at the start of a subsequent cycle should be based on nadir haematologic counts or maximum non-haematologic toxicity from the preceding cycle of therapy. Treatment may be delayed to allow sufficient time for recovery. Upon recovery patients should be retreated using the guidelines in **Tables 1, 2 and 3**, which are applicable for pemetrexed used as a single agent or in combination with cisplatin.

Table 1 - Dose modification table for Pemetrexed (as single agent or in combination) and cisplatin –Haematologic toxicities	
Nadir ANC < 500 /mm ³ and nadir platelets ≥ 50,000 /mm ³	75 % of previous dose (both pemetrexed and cisplatin)
Nadir platelets < 50,000 /mm ³ regardless of nadir ANC	75 % of previous dose (both pemetrexed and cisplatin)
Nadir platelets < 50,000 /mm ³ with bleeding ^a regardless of nadir ANC	50 % of previous dose (both pemetrexed and cisplatin)

^a These criteria meet the National Cancer Institute Common Toxicity Criteria (CTC v2.0; NCI 1998) definition of ≥CTC Grade 2 bleeding.

If patients develop non-haematologic toxicities ≥ Grade 3 (excluding neurotoxicity), pemetrexed should be withheld until resolution to less than or equal to the patient's pre-therapy value. Treatment should be resumed according to the guidelines in **Table 2**.

Table 2 - Dose modification table for Pemetrexed (as single agent or in combination) and cisplatin– Non-haematologic toxicities^{a, b}		
	Dose of Pemetrexed² (mg/m)	Dose for cisplatin² (mg/m)
Any Grade 3 or 4 toxicities except mucositis	75 % of previous dose	75 % of previous dose
Any diarrhoea requiring hospitalisation (irrespective of grade) or grade 3 or 4 diarrhoea.	75 % of previous dose	75 % of previous dose
Grade 3 or 4 mucositis	50 % of previous dose	100 % of previous dose

^a National Cancer Institute Common Toxicity Criteria (CTC v2.0; NCI 1998)

^b Excluding neurotoxicity

In the event of neurotoxicity, the recommended dose adjustment for pemetrexed and cisplatin is documented in **Table 3**. Patients should discontinue therapy if Grade 3 or 4 neurotoxicity is observed.

Table 3 - Dose modification table for Pemetrexed (as single agent or in combination) and cisplatin – Neurotoxicity		
CTC^a Grade	Dose of Pemetrexed (mg/m)²	Dose for cisplatin (mg/m)²
0 – 1	100 % of previous dose	100 % of previous dose
2	100 % of previous dose	50 % of previous dose

^a National Cancer Institute Common Toxicity Criteria (CTC v2.0; NCI 1998)

Treatment with Pemetrexed should be discontinued if a patient experiences any haematologic or non-haematologic Grade 3 or 4 toxicity after 2 dose reductions or immediately if Grade 3 or 4 neurotoxicity is observed.

Elderly: No dose reductions other than those recommended for all patients are necessary.

Paediatric population

There is no relevant use of Pemetrexed in the paediatric population in malignant pleural mesothelioma and non-small cell lung cancer.

Patients with renal impairment: Pemetrexed is primarily eliminated unchanged by renal excretion. Patients with creatinine clearance of ≥ 45 ml/min required no dose adjustments other than those recommended for all patients. The use of pemetrexed in patients with creatinine clearance below 45 ml/min is not recommended.

Patients with hepatic impairment: No relationships between AST (SGOT), ALT (SGPT), or total bilirubin and pemetrexed pharmacokinetics were identified. However, patients with hepatic impairment such as bilirubin > 1.5 times the upper limit of normal and/or aminotransferase > 3.0 times the upper limit of normal (hepatic metastases absent) or > 5.0 times the upper limit of normal (hepatic metastases present) have not been specifically studied.

Method of administration:

For Precautions to be taken before handling or administering Pemetrexed.

Special precautions for disposal and other handling

1. Use aseptic technique during the reconstitution and further dilution of pemetrexed for intravenous infusion administration.
2. Calculate the dose and the number of Pemetrexed vials needed. Each vial contains an excess of pemetrexed to facilitate delivery of label amount.
3. Reconstitute 100-mg vials with 4.2 ml of sodium chloride 9 mg/ml (0.9 %) solution for injection, without preservative, resulting in a solution containing 25 mg/ml pemetrexed. Reconstitute 500-mg vials with 20 ml of sodium chloride 9 mg/ml (0.9 %) solution for injection, without preservative, resulting in a solution containing 25 mg/ml pemetrexed. Gently swirl each vial until the powder is completely dissolved. The resulting solution is clear and ranges in colour from colourless to yellow or green-yellow without adversely affecting product quality. The pH of the reconstituted solution is between 6.6 and 7.8.

Further dilution is required.

1. The appropriate volume of reconstituted pemetrexed solution must be further diluted to 100 ml with sodium chloride 9 mg/ml (0.9 %) solution for injection, without preservative, and administered as an intravenous infusion over 10 minutes.
2. Pemetrexed infusion solutions prepared as directed above are compatible with polyvinyl chloride and polyolefin lined administration sets and infusion bags.
3. Parenteral medicinal products must be inspected visually for particulate matter and discolouration prior to administration. If particulate matter is observed, do not administer.
4. Pemetrexed solutions are for single use only. Any unused medicinal product or waste material must be disposed of in accordance with local requirements for cytotoxic medicinal products.

Preparation and administration precautions:

As with other potentially toxic anticancer agents, care should be exercised in the handling and

preparation of pemetrexed infusion solutions, especially by pregnant staff. The use of gloves is recommended. If a pemetrexed solution contacts the skin, wash the skin immediately and thoroughly with soap and water. If pemetrexed solutions contact the mucous membranes, flush thoroughly with water. Pemetrexed is not a vesicant. There is not a specific antidote for extravasation of pemetrexed. There have been few reported cases of pemetrexed extravasation, which were not assessed as serious by the investigator. Extravasation should be managed by local standard practice as with other non-vesicants.

Pemetrexed, a clear, colorless to yellow or yellow-green colored solution, once reconstituted; should be administered as an intravenous infusion over 10 minutes on the first day of each 21-day cycle.

4.3 Contraindications

- Hypersensitivity to the active substance or to any of the excipients.
- Breast-feeding.
- Concomitant yellow fever vaccine.

4.4 Special warnings and precautions for use

Pemetrexed can suppress bone marrow function as manifested by neutropenia, thrombocytopenia and anaemia (or pancytopenia) . Myelosuppression is usually the dose-limiting toxicity. Patients should be monitored for myelosuppression during therapy and pemetrexed should not be given to patients until absolute neutrophil count (ANC) returns to ≥ 1500 cells/mm³ and platelet count returns to $\geq 100,000$ cells/mm³. Dose reductions for subsequent cycles are based on nadir ANC, platelet count and maximum non-haematologic toxicity seen from the previous cycle.

Less toxicity and reduction in Grade 3/4 haematologic and non-haematologic toxicities such as neutropenia, febrile neutropenia and infection with Grade 3/4 neutropenia were reported when pre-treatment with folic acid and Vitamin B₁₂ was administered. Therefore, all patients treated with pemetrexed must be instructed to take folic acid and Vitamin B₁₂ as a prophylactic measure to reduce treatment-related toxicity.

Skin reactions have been reported in patients not pre-treated with a corticosteroid. Pre-treatment with dexamethasone (or equivalent) can reduce the incidence and severity of skin reactions.

The use of pemetrexed in patients with creatinine clearance of < 45 ml/min is not recommended.

Patients with mild to moderate renal insufficiency (creatinine clearance from 45 to 79 ml/min) should avoid taking non-steroidal anti-inflammatory drugs (NSAIDs) such as ibuprofen, and aspirin (> 1.3 g daily) for 2 days before, on the day of, and 2 days following pemetrexed administration. In patients with mild to moderate renal insufficiency eligible for pemetrexed therapy NSAIDs with long elimination half-lives should be interrupted for at least 5 days prior to, on the day of, and at least 2 days following pemetrexed administration.

Serious renal events, including acute renal failure, have been reported with pemetrexed alone or in association with other chemotherapeutic agents in patients whom had had underlying risk factors for the development of renal events including dehydration or pre-existing hypertension or diabetes.

The effect of third space fluid, such as pleural effusion or ascites, on pemetrexed is not fully

defined. Drainage of third space fluid collection prior to pemetrexed treatment should be considered, but may not be necessary.

Due to the gastrointestinal toxicity of pemetrexed given in combination with cisplatin, severe dehydration has been observed. Therefore, patients should receive adequate antiemetic treatment and appropriate hydration prior to and/or after receiving treatment.

Serious cardiovascular events, including myocardial infarction and cerebrovascular events have been uncommonly reported with pemetrexed, usually when given in combination with another cytotoxic agent in patients whom had had pre- existing cardiovascular risk factors.

Immunodepressed status is common in cancer patients. As a result, concomitant use of live attenuated vaccines is not recommended.

Pemetrexed can have genetically damaging effects. Sexually mature males are advised not to father a child during the treatment and up to 6 months thereafter. Contraceptive measures or abstinence are recommended. Owing to the possibility of pemetrexed treatment causing irreversible infertility, men are advised to seek counselling on sperm storage before starting treatment.

Women of childbearing potential must use effective contraception during treatment with pemetrexed.

Cases of radiation pneumonitis have been reported in patients treated with radiation either prior, during or subsequent to their pemetrexed therapy. Particular attention should be paid to these patients and caution exercised with use of other radiosensitising agents.

Nephrogenic diabetes insipidus and renal tubular necrosis were also reported in post marketing setting with pemetrexed alone or with other chemotherapeutic agents. Most of these events resolved after pemetrexed withdrawal. Patients should be regularly monitored for acute tubular necrosis, decreased renal function and signs and symptoms of nephrogenic diabetes insipidus (e.g. hypernatraemia).

Pemetrexed 100 mg

This medicinal product contains less than 1mmol (11 mg) of sodium per vial and is therefore considered essentially “sodium-free”.

Pemetrexed 500 mg

This medicinal product contains approximately 2.35 mmol (54 mg) sodium per vial. To be taken into consideration by patients on a controlled sodium diet.

4.5 Interaction with other medicinal products and other forms of interaction

Pemetrexed is mainly eliminated unchanged renally by tubular secretion and to a lesser extent by glomerular filtration. Concomitant administration of nephrotoxic drugs (e.g. aminoglycoside, loop diuretics, platinum compounds, cyclosporin) could potentially result in delayed clearance of pemetrexed. This combination should be used with caution. If necessary, creatinine clearance should be closely monitored.

Concomitant administration of substances that are also tubularly secreted (e.g. probenecid, penicillin) could potentially result in delayed clearance of pemetrexed. Caution should be made when these drugs are combined with pemetrexed. If necessary, creatinine clearance should be closely monitored.

In patients with normal renal function (creatinine clearance > 80 ml/min), high doses of non-

steroidal anti-inflammatory drugs (NSAIDs, such as ibuprofen > 1600 mg/day) and aspirin at higher dose (> 1.3 g daily) may decrease pemetrexed elimination and, consequently, increase the occurrence of pemetrexed adverse events. Therefore, caution should be made when administering higher doses of NSAIDs or aspirin, concurrently with pemetrexed to patients with normal function (creatinine clearance > 80 ml/min).

In patients with mild to moderate renal insufficiency (creatinine clearance from 45 to 79 ml/min), the concomitant administration of pemetrexed with NSAIDs (e.g. ibuprofen) or aspirin at higher dose should be avoided for 2 days before, on the day of, and 2 days following pemetrexed administration (see section 4.4).

Potential interaction with NSAIDs having longer half-lives such as piroxicam or rofecoxib is not known, therefore the concomitant administration with pemetrexed in patients with mild to moderate renal insufficiency should be interrupted for at least 5 days prior to, on the day of, and at least 2 days following pemetrexed administration. If concomitant administration of NSAIDs is necessary, patients should be monitored closely for toxicity, especially myelosuppression and gastrointestinal toxicity.

Pemetrexed undergoes limited hepatic metabolism. Therefore, pemetrexed would not be predicted to cause clinically significant inhibition of the metabolic clearance of drugs metabolised by CYP3A, CYP2D6, CYP2C9, and CYP1A2.

Interactions common to all cytotoxics:

Due to the increased thrombotic risk in patients with cancer, the use of anticoagulation treatment is frequent. The high intra-individual variability of the coagulation status during diseases and the possibility of interaction between oral anticoagulants and anticancer chemotherapy require increased frequency of INR (International Normalised Ratio) monitoring, if it is decided to treat the patient with oral anticoagulants.

Concomitant use contraindicated:

Yellow fever vaccine: risk of fatal generalised vaccinal disease.

Concomitant use not recommended:

Live attenuated vaccines (except yellow fever, for which concomitant use is contraindicated): risk of systemic, possibly fatal, disease. The risk is increased in who are already immunosuppressed by their underlying disease. Use an inactivated vaccine where it exists (poliomyelitis).

4.6 Fertility, pregnancy and lactation

Contraception in males and females

Women of childbearing potential must use effective contraception during treatment with pemetrexed. Pemetrexed can have genetically damaging effects. Sexually mature males are advised not to father a child during the treatment and up to 6 months thereafter. Contraceptive measures or abstinence are recommended.

Pregnancy

Pemetrexed, like other anti-metabolites, is suspected to cause serious birth defects when administered during pregnancy. Pemetrexed should not be used during pregnancy unless clearly necessary, after a careful consideration of the needs of the mother and the risk for the foetus.

Breastfeeding

It is not known whether pemetrexed is excreted in human milk and adverse reactions on the

suckling child cannot be excluded. Breast-feeding must be discontinued during pemetrexed therapy.

Fertility

Owing to the possibility of pemetrexed treatment causing irreversible infertility, men are advised to seek counselling on sperm storage before starting treatment.

4.7 Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been performed. However, pemetrexed may cause fatigue. Therefore patients should be cautioned against driving or operating machines if this event occurs.

4.8 Undesirable effects

Summary of the safety profile

The most commonly reported undesirable effects related to pemetrexed, whether used as monotherapy or in combination, are bone marrow suppression manifested as anaemia, neutropenia, leukopenia, thrombocytopenia; and gastrointestinal toxicities, manifested as anorexia, nausea, vomiting, diarrhoea, constipation, pharyngitis, mucositis, and stomatitis. Other undesirable effects include renal toxicities, increased aminotransferases, alopecia, fatigue, dehydration, rash, infection/sepsis and neuropathy. Rarely seen events include Stevens-Johnson syndrome and Toxic epidermal necrolysis.

Tabulated list of adverse reactions

The table below provides the frequency and severity of undesirable effects in patients with mesothelioma who receive cisplatin and pemetrexed.

Adverse reactions

Frequency estimate: Very common, Common, Uncommon, Rare, Very rare and not known.

Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

System Organ class	Frequency	Event*
Blood and lymphatic system disorders	Very common	Neutrophils/ Granulocytes decreased
		Leukocytes decreased
		Haemoglobin decreased
		Platelets decreased
Metabolism and nutrition disorders	Common	Dehydration
Nervous system disorders	Very common	Neuropathy- Sensory
	Common	Taste disturbance
Eye disorders	Common	Conjunctivitis
Gastrointestinal disorders	Very common	Diarrhoea
		Vomiting

		Stomatitis/ Pharyngitis
		Nausea
		Anorexia
		Constipation
	Common	Dyspepsia
Skin and subcutaneous tissue disorders	Very common	Rash
		Alopecia
Renal and urinary disorders	Very common	Creatinine elevation
		Creatinine clearance decreased**
General disorders and administration site conditions	Very common	Fatigue

* Refer to National Cancer Institute CTC version 2 for each grade of toxicity except the term “creatinine clearance decreased”

** which is derived from the term “renal/genitourinary other”.

The table below provides the frequency and severity of undesirable effects in patients receiving single agent pemetrexed with folic acid and Vitamin B12 supplementation. All patients were diagnosed with locally advanced or metastatic non-small cell lung cancer and received prior chemotherapy.

System organclass	Frequency	Event*
Blood and lymphatic system disorders	Very Common	Neutrophils/ Granulocytes decreased
		Leukocytes decreased
		Haemoglobin decreased
	Common	Platelets decreased
Gastrointestinal disorders	Very Common	Diarrhoea
		Vomiting
		Stomatitis/Pharyngitis
		Nausea
		Anorexia
	Common	Constipation
Hepatobiliary disorders	Common	SGPT (ALT) elevation
		SGOT (AST) elevation
Skin and subcutaneous tissue disorders	Very Common	Rash/ desquamation
	Common	Pruritus
		Alopecia
General disorders and administration site conditions	Very Common	Fatigue
	Common	Fever

*Refer to National Cancer Institute CTC version 2 for each grade of toxicity.

The table below provides the frequency and severity of undesirable effects considered in patients with NSCLC who receive cisplatin and pemetrexed. All patients received therapy as initial treatment for locally advanced or metastatic NSCLC and patients were fully supplemented with folic acid and Vitamin B₁₂.

System organ class	Frequency	Event**
Blood and lymphatic system disorders	Very common	Hemoglobin decreased
		Neutrophils/ Granulocytes decreased
		Leukocytes Decreased
		Platelets Decreased
Nervous system disorders	Common	Neuropathy-sensory
		Taste disturbance
Gastrointestinal disorders	Very common	Nausea
		Vomiting
		Anorexia
		Constipation
		Stomatitis/ Pharyngitis
		Diarrhoea without colostomy
	Common	Dyspepsia/ Heartburn
Skin and subcutaneous tissue disorders	Very common	Alopecia
	Common	Rash/desquamation
Renal and urinary disorders	Very common	Creatinine elevation
General disorders and administration site conditions	Very common	Fatigue

**Refer to National Cancer Institute CTC (v2.0; NCI 1998) for each Grade of Toxicity.

The table below provides the frequency and severity of undesirable effects in patients receiving single agent pemetrexed. All patients were diagnosed with Stage IIIB or IV NSCLC and had received prior platinum-based chemotherapy. Patients were fully supplemented with folic acid and Vitamin B₁₂.

System organ class	Frequency	Event**
Blood and lymphatic system disorders	Very common	Hemoglobin decreased
	Common	Leukocytes decreased
		Neutrophils decreased
Nervous system disorders	Common	Neuropathy- sensory
Gastrointestinal disorders	Very common	Nausea
		Anorexia
	Common	Vomiting

		Mucositis/stomatitis
Hepatobiliary disorders	Common	ALT (SGPT) elevation
		AST (SGOT) elevation
Skin and subcutaneous tissue disorders	Common	Rash/desquamation
General disorders and administration site disorders	Very common	Fatigue
	Common	Pain
		Edema
Renal Disorders	Common	Renal disorders*****

Common Terminology Criteria for Adverse Event; NCI = National Cancer Institute; SGOT = serum glutamic oxaloacetic aminotransferase; SGPT = serum glutamic pyruvic aminotransferase.

** Refer to NCI CTCAE Criteria (Version 3.0; NCI 2003) for each grade of toxicity. The reporting rates shown are according to CTCAE version 3.0.

***** Combined term includes increased serum/blood creatinine, decreased glomerular filtration rate, renal failure and renal/genitourinary- other.

Serious cardiovascular and cerebrovascular events, including myocardial infarction, angina pectoris, cerebrovascular accident and transient ischaemic attack have been uncommonly reported with pemetrexed, usually when given in combination with another cytotoxic agent. Most of the patients in whom these events have been observed had pre-existing cardiovascular risk factors.

Rare cases of hepatitis, potentially serious, have been reported with pemetrexed.

Pancytopenia has been uncommonly reported with pemetrexed.

Cases of colitis (including intestinal and rectal bleeding, sometimes fatal, intestinal perforation, intestinal necrosis and typhlitis) have been reported uncommonly in patients treated with pemetrexed.

Cases of interstitial pneumonitis with respiratory insufficiency, sometimes fatal, have been reported uncommonly in patients treated with pemetrexed.

Uncommon cases of oedema have been reported in patients treated with pemetrexed.

Oesophagitis/ radiation oesophagitis has been uncommonly reported with pemetrexed.

Sepsis, sometimes fatal, has been commonly reported with pemetrexed.

During post marketing surveillance, the following adverse reactions have been reported in patients treated with pemetrexed:

Uncommon cases of acute renal failure have been reported with pemetrexed alone or in association with other chemotherapeutic agents.

Uncommon cases of radiation pneumonitis have been reported in patients treated with radiation either prior, during or subsequent to their pemetrexed therapy.

Rare cases of radiation recall have been reported in patients who have received radiotherapy previously.

Uncommon cases of peripheral ischaemia leading sometimes to extremity necrosis have been reported.

Rare cases of bullous conditions have been reported including Stevens-Johnson syndrome and

Toxic epidermal necrolysis which in some cases were fatal.

Rarely, immune-mediated haemolytic anaemia has been reported in patients treated with pemetrexed.

Rare cases of anaphylactic shock have been reported.

Nephrogenic diabetes insipidus and renal tubular necrosis have been reported in post marketing setting with an unknown frequency

4.9 Overdose

Symptoms of overdose include neutropenia, anaemia, thrombocytopenia, mucositis, sensory polyneuropathy and rash. Anticipated complications of overdose include bone marrow suppression as manifested by neutropenia, thrombocytopenia and anaemia. In addition, infection with or without fever, diarrhoea, and/or mucositis may be seen. In the event of suspected overdose, patients should be monitored with blood counts and should receive supportive therapy as necessary. The use of calcium folinate / folinic acid in the management of pemetrexed overdose should be considered.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Folic acid analogues, ATC code: L01BA04

Mechanism of action

Pemetrexed (pemetrexed) is a multi-targeted anti-cancer antifolate agent that exerts its action by disrupting crucial folate-dependent metabolic processes essential for cell replication.

Pharmacodynamic effects

Pemetrexed behaves as a multitargeted antifolate by inhibiting thymidylate synthase (TS), dihydrofolate reductase (DHFR), and glycinamide ribonucleotide formyltransferase (GARFT), which are key folate-dependent enzymes for the *de novo* biosynthesis of thymidine and purine nucleotides. Pemetrexed is transported into cells by both the reduced folate carrier and membrane folate binding protein transport systems. Once in the cell, pemetrexed is rapidly and efficiently converted to polyglutamate forms by the enzyme folylpolyglutamate synthetase. The polyglutamate forms are retained in cells and are even more potent inhibitors of TS and GARFT. Polyglutamation is a time- and concentration-dependent process that occurs in tumour cells and, to a lesser extent, in normal tissues. Polyglutamated metabolites have an increased intracellular half-life resulting in prolonged drug action in malignant cells.

5.2 Pharmacokinetic properties

Pemetrexed has a steady-state volume of distribution of 9 l/m². Pemetrexed is approximately 81 % bound to plasma proteins. Binding was not notably affected by varying degrees of renal impairment. Pemetrexed undergoes limited hepatic metabolism. Pemetrexed is primarily eliminated in the urine, with 70 % to 90 % of the administered dose being recovered unchanged in urine within the first 24 hours following administration. Pemetrexed is actively secreted by OAT3 (organic anion transporter). Pemetrexed total systemic clearance is 91.8 ml/min and the elimination half-life from plasma is 3.5 hours in patients with normal renal function (creatinine clearance of 90 ml/min). Between patient variability in clearance is moderate at 19.3 %. Pemetrexed total systemic exposure (AUC) and maximum plasma

concentration increase proportionally with dose. The pharmacokinetics of pemetrexed are consistent over multiple treatment cycles.

The pharmacokinetic properties of pemetrexed are not influenced by concurrently administered cisplatin. Oral folic acid and intramuscular Vitamin B₁₂ supplementation do not affect the pharmacokinetics of pemetrexed.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Mannitol

Hydrochloric acid (*for pH adjustment*)

Sodium hydroxide (*for pH adjustment*)

6.2 Incompatibilities

Pemetrexed is physically incompatible with diluents containing calcium, including lactated Ringer's injection and Ringer's injection. In the absence of other compatibility studies this medicinal product must not be mixed with other medicinal products.

6.3 Shelf life

Unopened vial: 3 years

Reconstituted and infusion solutions:

When prepared as directed, reconstituted and further diluted infusion solutions of Pemetrexed contain no antimicrobial preservatives.

From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would not be longer than 24 hours at 2°C to 8°C.

6.4 Special precautions for storage

Unopened vial

Store at < 30°C before reconstituted.

For storage conditions after reconstitution of the medicinal product, see section 6.3.

6.5 Nature and contents of container

Type I glass vial (10 R) with rubber chlorobutyl or bromobutyl stopper with a green Flip-off seal containing 100 mg of pemetrexed.

Type I glass vial (50 H) with rubber chlorobutyl or bromobutyl stopper with an orange Flip-off seal containing 500 mg of pemetrexed.

Pack of 1 vial.

MANUFACTURER

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DATE OF REVISION OF THE TEXT

05/04/2021